

بحث الكور 505

بحث الكور 505 الخاص بالـ WCF

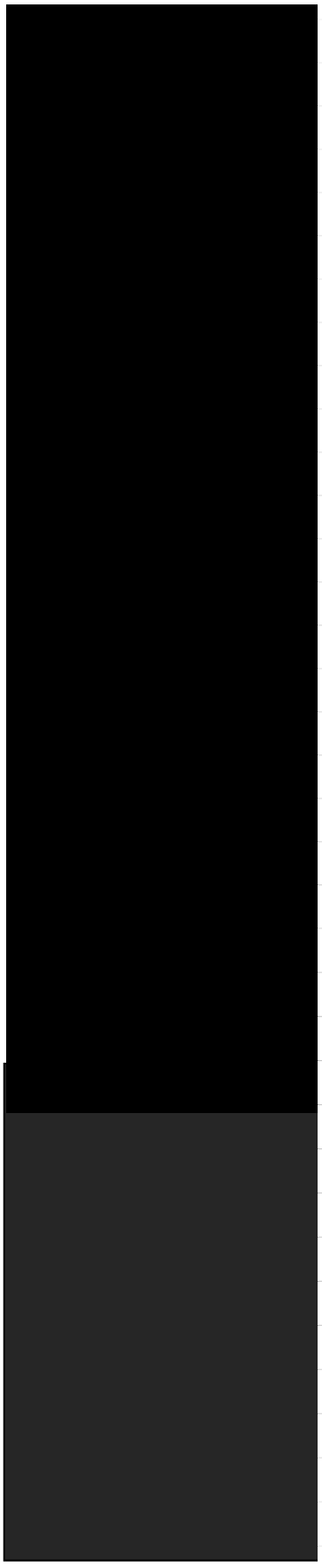
كريم رفعت

2016/03/10

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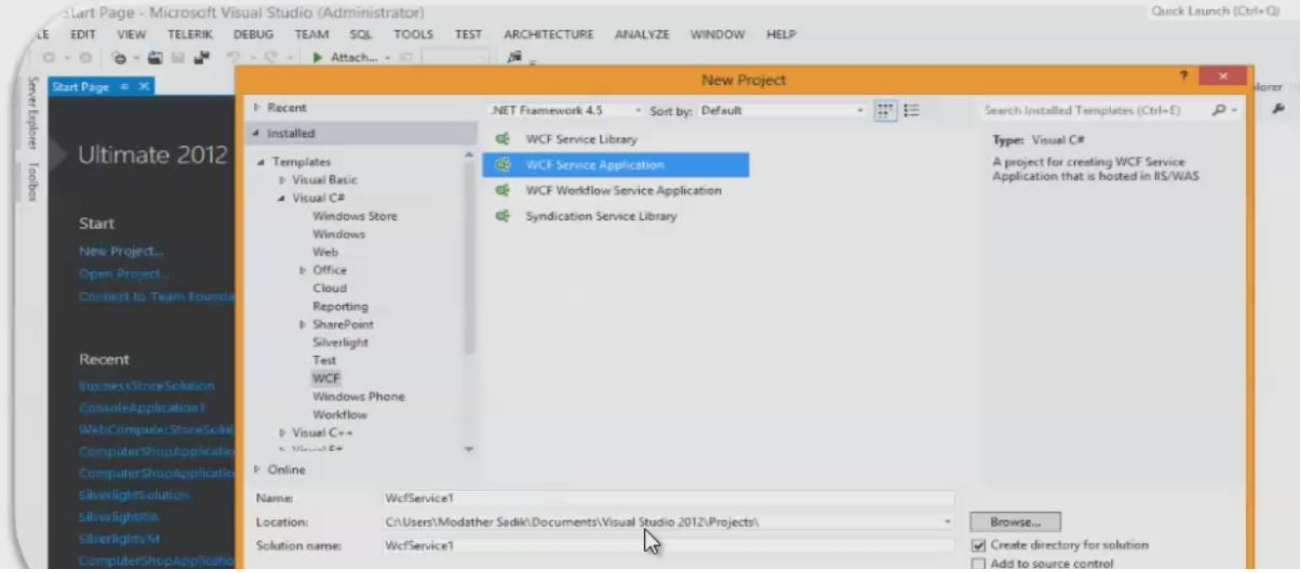


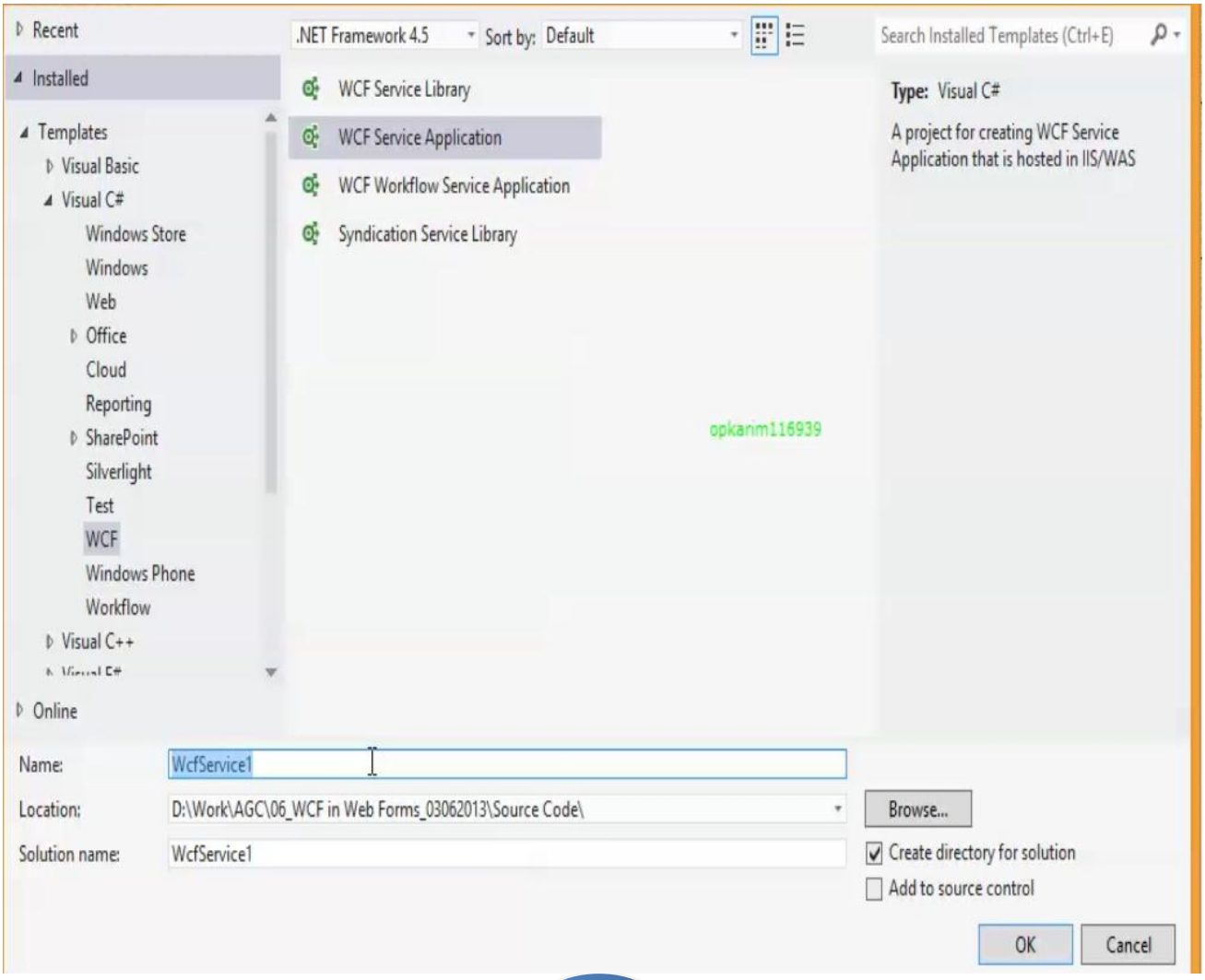
WCF4 USING C#

CONTENTS

- What's **WCF** ? & What's service ?
- What does **WCF** solve ?
- **WCF** ABCs ?
  - **A**ddress (Where?)
  - **B**inding (How?)
  - **C**ontract (What?)
- Endpoint

CREATING FIRST **WCF** APPLICATION





PROJECT FILES

- IService.cs
- Service.svc
- Web.config

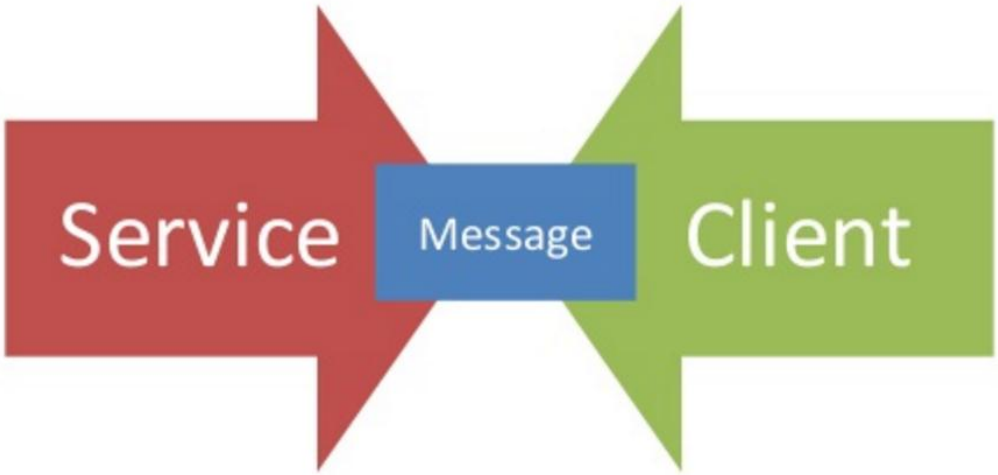
What definition you get on web ?

Service Oriented programming model to develop connected applications

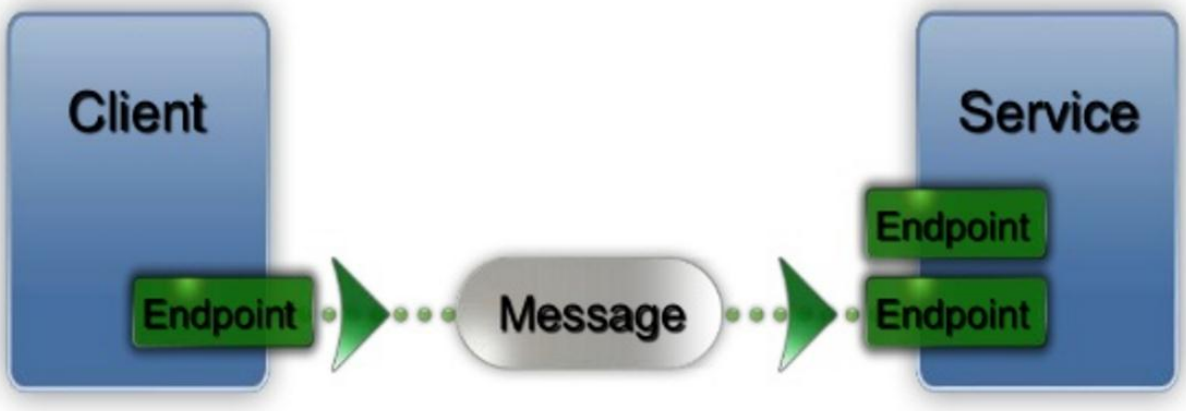
SDK to develop and deploy services on windows

Unifies the existing suite of .Net distributed technologies into a single programming model.

Services and Clients



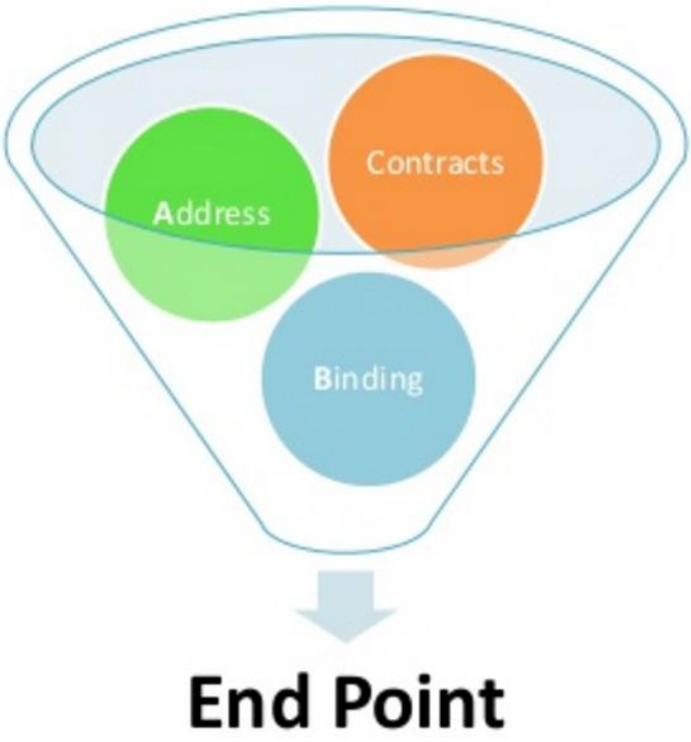
Endpoints



End Points

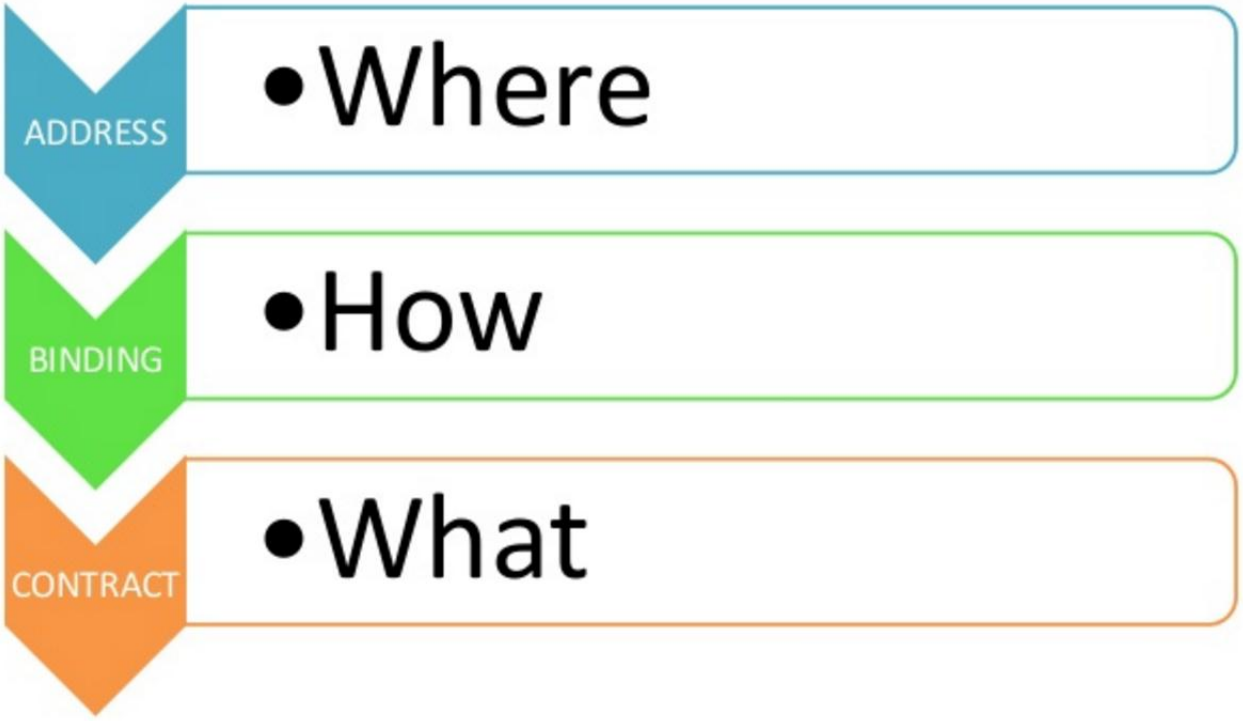


End Points



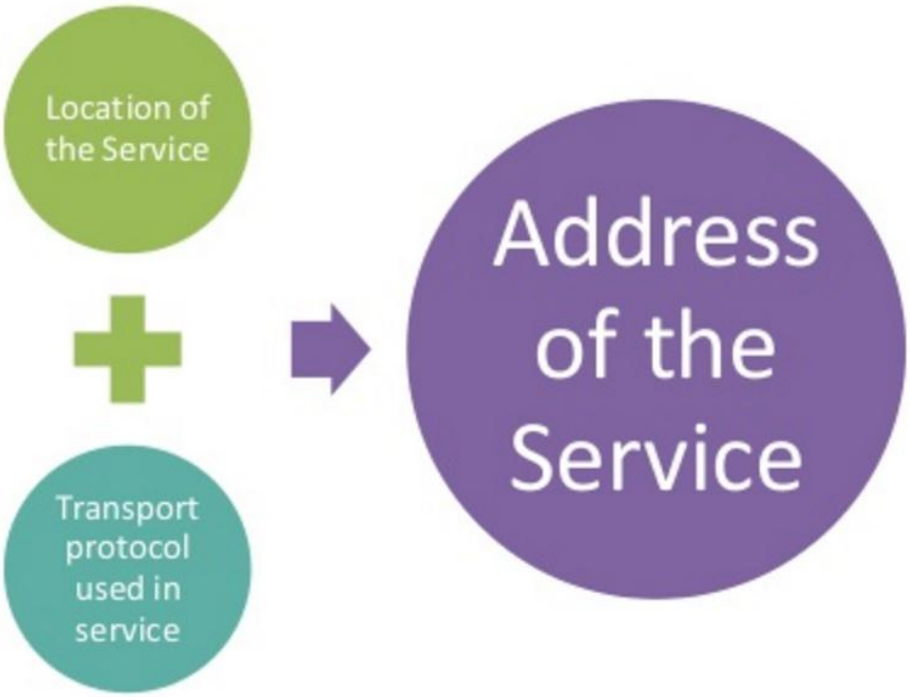
End Point

End Point



Address (A)

Every service is associated with a unique address.



Address (A)

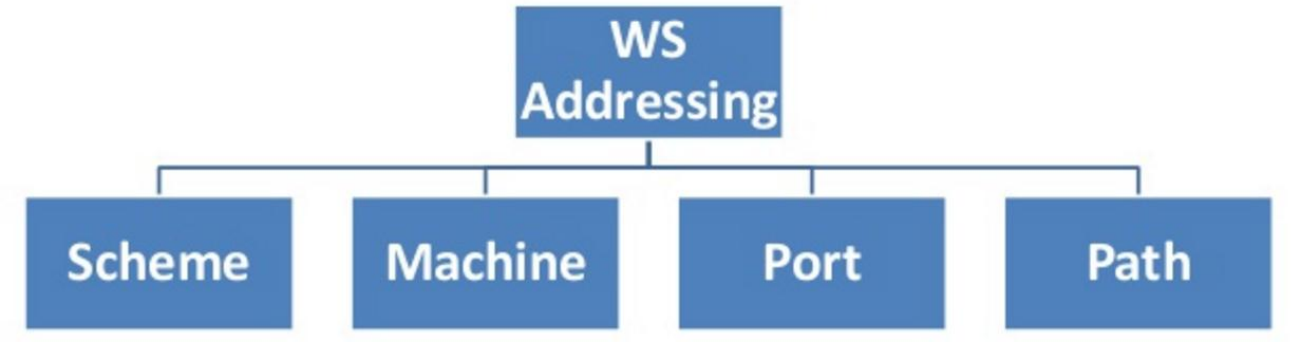
Address specifies where the service is residing.

This is an URL(Uniform Resource Locator).

Address URL identifies , location of the service

Address should follow the Web Service Addressing(WS-Addressing) standard.

Address (A)



Address (A)

Scheme	• This is top level portion of the address. This is not as same as of protocols.
Machine	• This identifies the machine name. This can be a public URL or a local identifier
Port	• This is port number. This is an optional part.
Path	• This is used to locate the path. This gives path of the service.

Address (A)

HTTP	• http://localhost:8001/Dell • http://localhost/Dell
TCP	• net.tcp://localhost:800/Dell • net.tcp://localhost/Dell
MSMQ	• net.msmq/private/MyService • Net.msmq/://localhost/Dell
IPC	• net.pipe://localhost/MyPipe
Peer network	• Net.pipe://localhost.MyService

Binding (B)

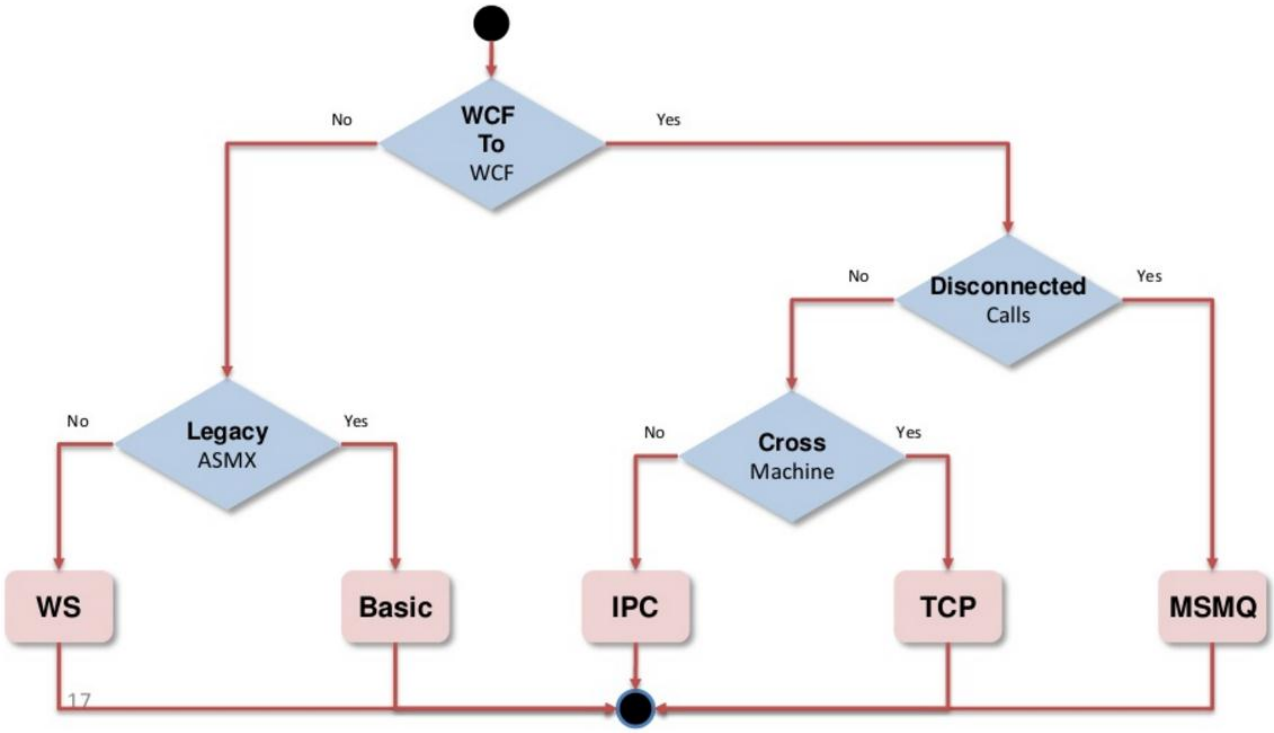
- Describes how a service communicates.
- This specifies which Protocol to be used.
- This specifies the encoding method to format the message content.
- This specifies the Security requirements
- This specifies the message exchange format.
- This specifies message session mode.
- Developer could create custom Binding also.

Binding (B)

- Transport Protocol
- Message Encoding
- Communication Pattern
- Security
- Transaction Property
- Inter Operability

Binding (B)

Choosing Binding



Binding (B)

Binding Classes

Binding Name as of class	Transport	Encoding	Interoperable
BasicHttpBinding	HTTP/HTTPS	Text,MTOM	Yes
NetTcpBinding	TCP	Binary	No
NetPeerBinding	P2P	Binary	No
NetNamedPipeBinding	IPC	Binary	No
WSHttpBinding	HTTP/HTTPS	Text,MTOM	Yes
WSFederationHttpBinding	HTTP/HTTPS	Text,MTOM	Yes
WSDualHttpBinding	HTTP	Text,MTOM	Yes
NetMsmqBinding	MSMQ	Binary	No
MsmqIntegrationBinding	MSMQ	Binary	Yes

Binding (B)

Configuring Binding

```
<endpoint address="Calculator"
          bindingSectionName="basicProfileBinding"
          bindingConfiguration="Binding1"
          contractType="ICalculator" />
```

```
<bindings>
  <basicProfileBinding>
    <binding configurationName="Binding1"
             hostnameComparisonMode="StrongWildcard"
             transferTimeout="00:10:00"
             maxMessageSize="65536"
             messageEncoding="Text"
             textEncoding="utf-8"
    </binding>
  </basicProfileBinding>
</bindings>
```

Service Contract (C)

A WCF Contract is a collection of Operations

All WCF services exposes contract.

This is a platform neutral and standard way to say , what the service will do.

Defines , what a Service communicates.

CONTRACTS

- Data Contract
- Message Contract
- Service Contract
- Operation Contract

Contract (C)

Types of Contracts

Service Contract
- Describes which operation client can perform on the services.

Data Contract
- Defines which Data Type are passed to and from the services. It provides built-in contract for implicit type.

Fault Contract
- Which error raise by service and how service propagates and handles error to its client.

Message Contract
- Allow the service to interact directly with the message . Message contract can be typed or un typed. This can be used for interoperability.

Contract (C)

Service Contract

A Service Contract reflects specific business

Describe which operations client can perform

Maps CLR types to WSDL

Contract (C)

Service Contract

Interfaces are used to explicitly define a Contract

Classes may also be used to define a Contract.

[ServiceContract] attribute is being used by interface/class to qualify them as a contract.

ServiceContract are implicitly public.

Contract (C)

Service Contract

```
// Define a service contract.
[ServiceContract(Namespace="http://Microsoft.ServiceModel.Samples")]
public interface IDataContractCalculator
{
    [OperationContract]
    ComplexNumber Add(ComplexNumber n1, ComplexNumber n2);
    [OperationContract]
    ComplexNumber Subtract(ComplexNumber n1, ComplexNumber n2);
    [OperationContract]
    ComplexNumber Multiply(ComplexNumber n1, ComplexNumber n2);
    [OperationContract]
    ComplexNumber Divide(ComplexNumber n1, ComplexNumber n2);
}
```

Contract (C)

Data Contract

These are the contractual agreement about the format and structure of the payload data in message exchange between a service and its consumer.

Defines which Data types are passed to and from the service.

Its specifies CLR type to XML schema.

Contract (C)

Data Contract

DataContract are preferred WCF way to enable Serialization.

WCF defines implicit contract for built in types like int and string.

Developer has to explicitly expose complex types as Data Contract.

[DataContract] and [DataMember] attribute are used to define a type as Data Contract.

Contract (C)

Data Contract

```
[DataContract]
public class ComplexNumber
{
    [DataMember]
    public double Real = 0.0D;
    [DataMember]
    public double Imaginary = 0.0D;
}
```

Contract (C)

Message Contract

It gives control over SOAP message structure on both header and body content.

Developer can designate optional SOAP headers.

It provides additional control over the WSDL generation.

It is used to interoperate with another non- WCF service.

It is used to control security issue at level of message.

Contract (C)

Fault Contract

This translates .Net Exception to SOAP fault propagated to consumer

This can be applied to operation only.

This is not Inheritable.

This can be applied multiple times.

This enables developer to declare which faults a given service operation might issue if things goes wrong.

Creating End Point

Could be created declaratively in configuration file.

Could be created imperatively through code.

Any thing done through code can be done with configuration file and vice versa.

It is considered good practice to use configuration file to specify End points. This accommodates changes without recompiling the code.

Defining Endpoints

```
<?xml version="1.0" encoding="utf-8" ?>
<configuration xmlns="http://schemas.microsoft.com/.NetConfiguration/v2.0">
  <system.serviceModel>
    <services>
      <service serviceType="CalculatorService">
        <endpoint address="Calculator"
                  bindingSectionName="basicProfileBinding"
                  contractType="ICalculator" />
      </service>
    </services>
  </system.serviceModel>
</configuration>
```

CHANGE STANDARD ENDPOINT

```
1.  <system.serviceModel>
2.      <services>
3.        <service name="TService">
4.          <endpoint kind="webHttpEndpoint" contract="ITService"
5.          endpointConfiguration="TConfig"/>
6.        </service>
7.      </services>
8.      <standardEndpoints>
9.        <webHttpEndpoint>
10.         <standardEndpoint name="TConfig" transferMode="Buffered"/>
11.       </webHttpEndpoint>
12.     </standardEndpoints>
13. </system.serviceModel>
```

Multiple End Point

For service exposed to multiple clients ; it makes sense to specify more than one End point.

This enables client to use the endpoint that is most applicable for them.

When creating multiple endpoints each client must have unique address

If two client uses the same address , error will raise at service load time.

Multiple End Point

- <endpoint address="http://localhost:8890/a" binding="wsHttpBinding" contract="HotsingSamples.IService1"/>
- <endpoint address="http://localhost:8000/b" binding="basicHttpBinding" contract="HotsingSamples.IService1"/>
- <endpoint address="net.tcp://localhost:8001/c" binding="netTcpBinding" contract="HotsingSamples.IService1"/>

Hosting

WCF services can not exist in void.

It must be hosted.

WCF services are hosted in windows process called host process.

A single host process can host multiple service.

A same service can be hosted in multiple host process.

WCF basic task cycle